

Firewalls and IDS/IPS — Guided Notes ANSWER KEY

Unit 4.2.1 | CompTIA Network+ N10-009 Obj. 4.3 | N10-008 4.3

For instructor use / distribute after students complete the notes.

Question 1

A stateless firewall evaluates each packet _____, while a stateful firewall maintains a _____ table that tracks active connections. Why does stateless filtering require explicit rules for return traffic?

Answer: In isolation (independently); state/connection. Stateless firewalls have no memory of prior packets, so the return traffic from an outbound session looks like a new unsolicited inbound connection and must be explicitly permitted.

Question 2

A next-generation firewall (NGFW) differs from a traditional stateful firewall by inspecting the _____ layer, allowing policy decisions based on _____ identity rather than just port number.

Answer: Application; application (e.g., block BitTorrent even if it runs on port 80).

Question 3

Firewalls process rules _____. If a broad deny rule appears above a specific permit rule, the permit rule will _____ be reached. Best practice is to place _____ rules at the top.

Answer: Top-down (first match wins); never; most specific.

Question 4

An implicit deny is a rule at the _____ of the firewall rule set that blocks all traffic not matched by a preceding rule. Why is it important to _____ this rule?

Answer: Bottom; log. Logging the implicit deny reveals misconfigured traffic (legitimate flows that need a rule) and attack traffic (probing or scanning).

Question 5

A DMZ (demilitarized zone) places public-facing servers on a _____ network segment between the internet and the internal network. If a DMZ web server is compromised, what prevents the attacker from immediately reaching internal systems?

Answer: Separate (isolated); firewall rules between the DMZ and the inside network limit what DMZ servers can initiate toward the internal zone — the attacker hits a second firewall barrier.

Question 6

An IDS is _____ — it receives a copy of traffic and generates alerts. An IPS is _____ — traffic passes through it and it can actively _____ malicious packets. What happens to network traffic if an inline IPS fails in fail-closed mode?

Answer: Passive; inline; drop/block. In fail-closed mode, traffic stops entirely — the network goes down but stays secure.

Question 7

Signature-based detection matches traffic against _____ attack patterns, so it cannot detect _____. Anomaly-based detection flags deviations from a _____ but produces more _____ than signature-based systems.

Answer: Known; zero-day (novel) attacks; baseline; false positives.

Question 8 — Real World Application

REAL WORLD: A network admin adds a firewall rule permitting guest VLAN traffic to the student portal on port 443, but the portal remains unreachable from the guest network. The rule is syntactically correct. What is the most likely cause, and how do you verify it?

Answer: Rule order — a broader deny rule for the guest VLAN appears above the new permit rule, so the firewall matches it first and stops processing. Verify by reviewing the full rule set top-to-bottom and using the firewall's packet-trace or log function to see which rule is matching the traffic.